

Logupload Synchronization

Reboot Sequence Sync for dcm-agent

The Problem

Four independent subsystems read `/opt/logs/PreviousLogs/` after reboot — with no coordination:

- `backup_logs` — moves logs to `PreviousLogs/` directory
- `update-prev-reboot-info` — derives and persists reboot reason
- `uploadstblogs` — archives and uploads logs to a server
- `telemetry2_0` — grep-scans `PreviousLogs/` for telemetry markers (one-shot)

What goes wrong?

- Logs read before backup completion
- Reboot reasons derived from incomplete logs
- Uploads with incorrect timestamps or missing data
- Telemetry markers permanently lost — one-shot report, never retried

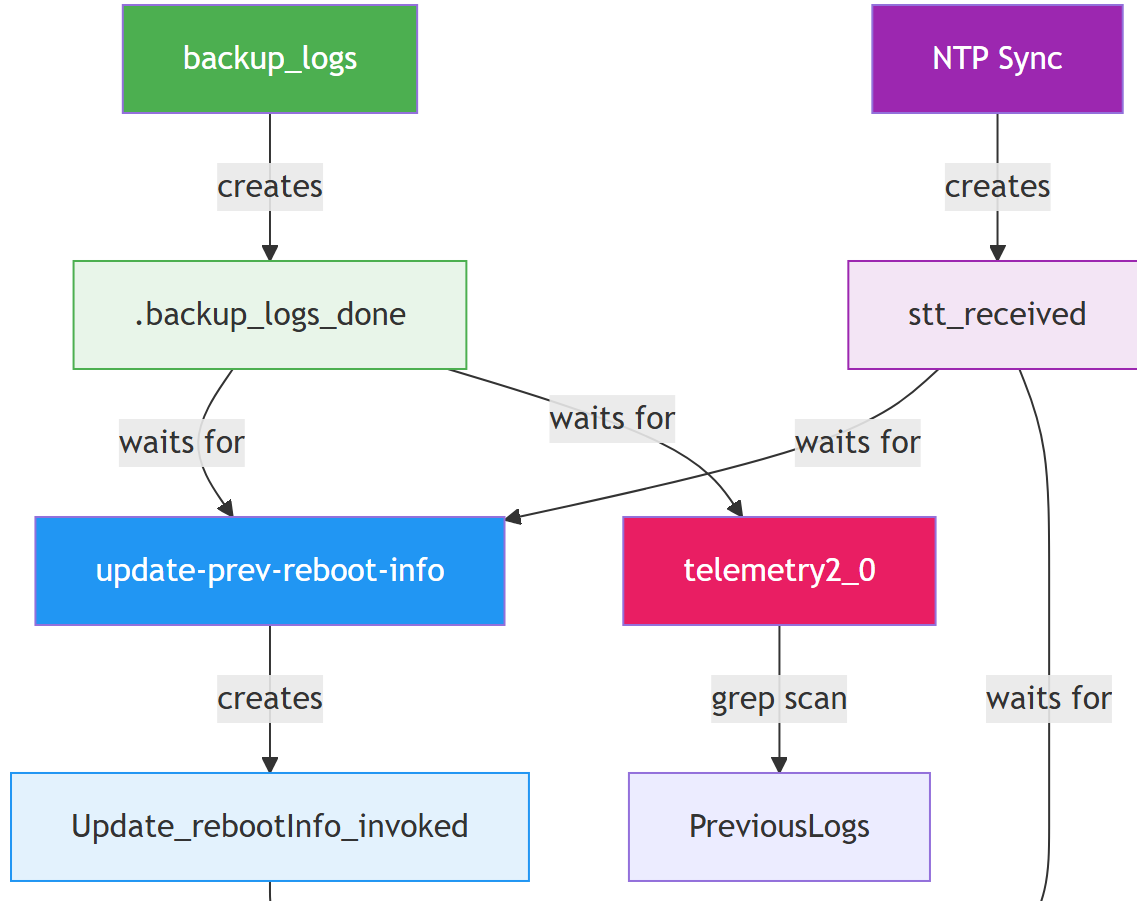
Current State

- Four subsystems launch independently at boot
- No ordering guarantees
- Race conditions between log backup, reboot-info, upload, and telemetry
- Timestamps may be inaccurate (no NTP sync)
- Telemetry scans PreviousLogs/ before files are moved

Target State

- Sentinel file chain enforces strict order
- Each subsystem waits for prerequisites
- NTP synchronization required before upload
- Telemetry polls `.backup_logs_done` before grep scan
- Clean skip on timeout / failure

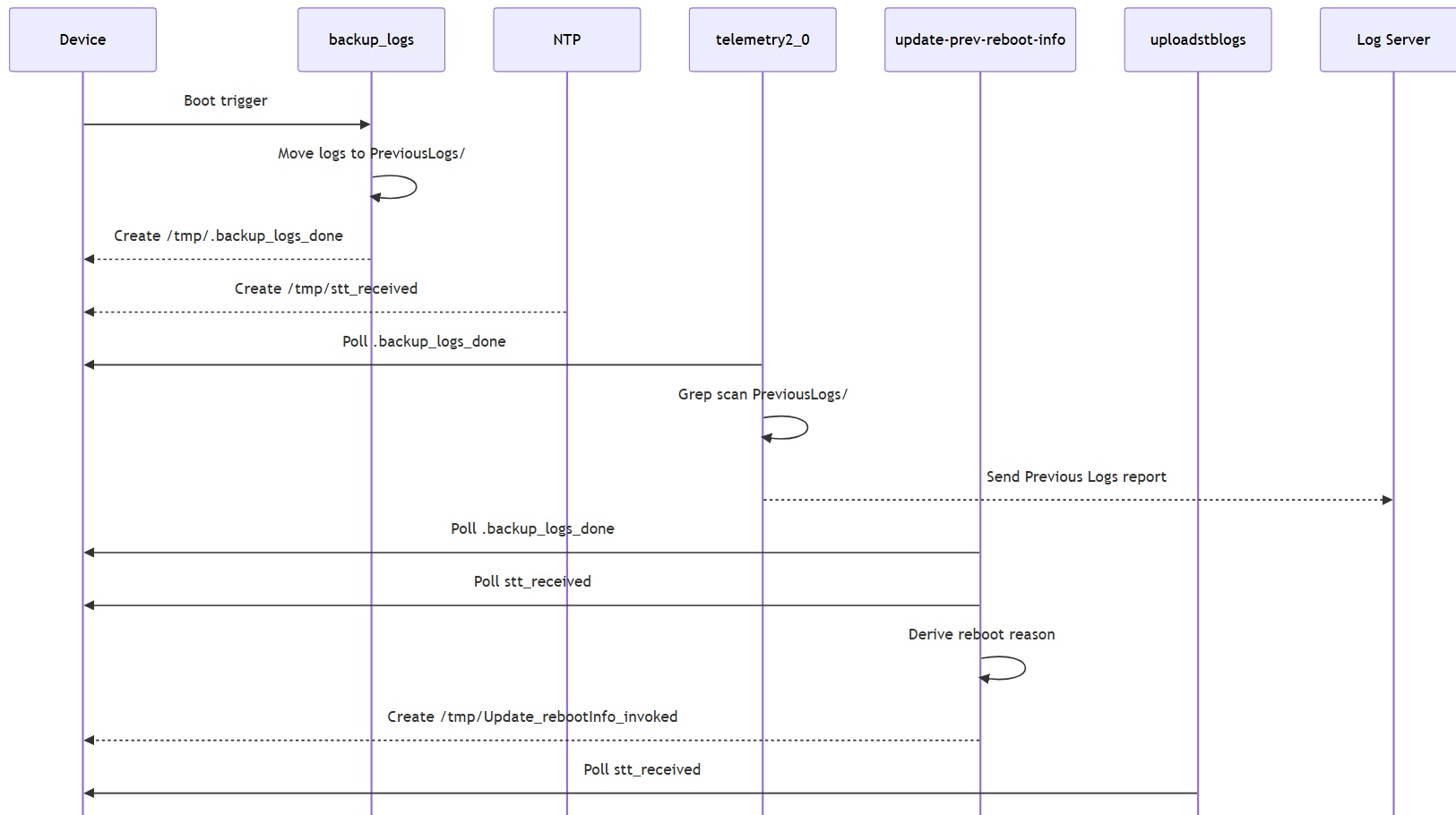
Sentinel File Protocol



Execution Flow

1. Device Reboots
2. `backup_logs` starts → moves logs to `/opt/logs/PreviousLogs/`
3. On success → creates `/tmp/.backup_logs_done`
4. `update-prev-reboot-info` polls for:
 - `/tmp/.backup_logs_done`
 - `/tmp/stt_received` (NTP sync)
5. Derives reboot reason → creates `/tmp/Update_rebootInfo_invoked`
6. `uploadstblogs` polls for:
 - `/tmp/stt_received`
 - `/tmp/Update_rebootInfo_invoked`
7. Proceeds with log upload

Execution Sequence



Key Components

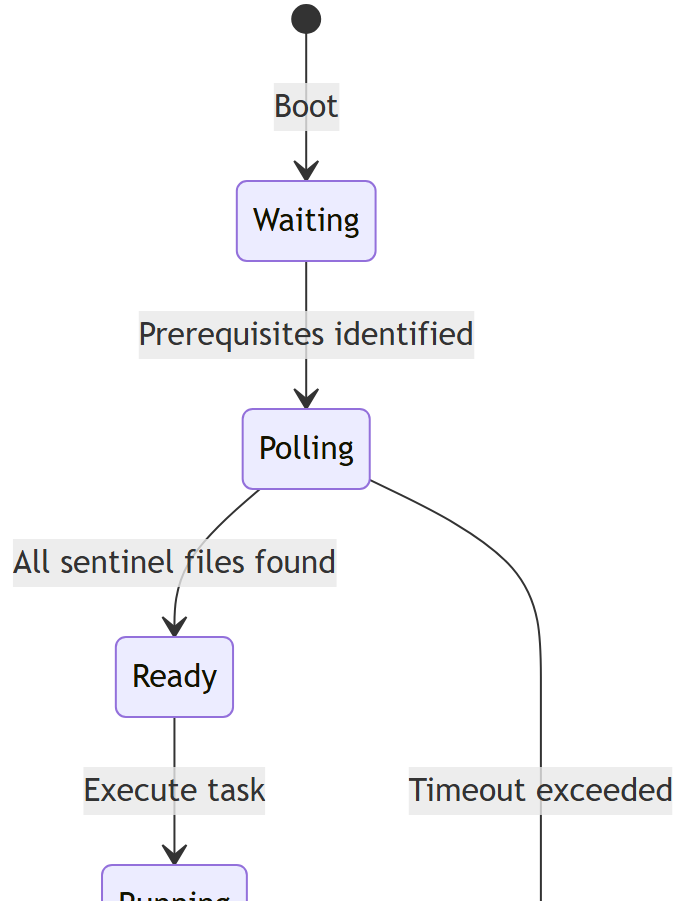
Component	Language	Role
backup_logs	C (migrated from shell)	Move logs to backup directory
Reboot Manager	C	Derive and persist reboot reason
uploadstblogs	C	Archive & upload logs to server
telemetry2_0	C	Grep-scan PreviousLogs/ for markers
Sentinel Files	Filesystem	Inter-process synchronization

Error Handling

- Each subsystem polls with a **configurable timeout**
- If a sentinel file is not created in time:
 - Dependent subsystem logs an error
 - **Skips its operation cleanly**
- No cascading failures — each stage fails independently
- All failures are **logged** for diagnostics

Component	Timeout	Action on timeout
update-prev-reboot-info	60s	Exit <code>ERROR_GENERAL</code>
telemetry2_0	60s	Skip PreviousLogs report, use current logs
uploadstblogs	120s	Skip reboot log upload entirely

Subsystem State Machine



backup_logs Migration

Migrated from shell script → C for:

- Better performance on resource-constrained devices
- Modular architecture:
 - `config_manager` — configuration handling
 - `backup_engine` — core log backup logic
 - `special_files` — special file handling
 - `sys_integration` — system integration layer
- Comprehensive unit tests with Google Test

Design Principles

- No dynamic memory allocation — memory pools where needed
- Platform-neutral — portable across embedded targets
- Thread-safe — safe concurrent access
- Fixed-point arithmetic — no floating-point dependency
- Secure coding — input validation, no buffer overflows
- Modular — each subsystem independently testable

Cross-Repo Interface Contract

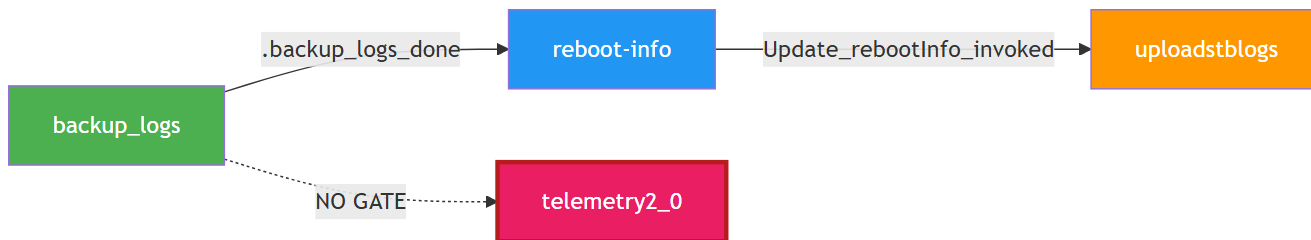
The sentinel `/tmp/.backup_logs_done` is a 3-repo interface:

Repo	Role	Action
dcm-agent	Writer	<code>backup_logs</code> creates sentinel on success
reboot-manager	Consumer	Polls before reading PreviousLogs/
telemetry	Consumer	Polls before grep-scanning PreviousLogs/

- All path changes must be **coordinated across all three repos**
- Sentinel resides in `/tmp/` — volatile, cleared on every reboot
- Telemetry fix tracked as a **separate change** in the telemetry repo
- The write side is already planned (TASK-B1 in reboot-sequence-sync)

The Telemetry Gap

Discovery: `telemetry2_0` reads `PreviousLogs/` at boot — outside the sentinel chain



- `PERSIST_LOG_MON_REF` enabled on all builds
- `PreviousLogs` grep report is **fire-and-forget** — never retried
- If it reads incomplete data, telemetry markers are **permanently lost**
- Fix: poll `.backup_logs_done` in telemetry repo before `PreviousLogs` scan

Summary

Sentinel file chain eliminates race conditions at boot

backup_logs → → uploadstblogs

3-repo coordination: dcm-agent, reboot-manager, telemetry

Reliable, ordered, and fault-tolerant log processing

Thank You

dcm-agent Repository
